

X2UJ: From AmbiX to UJ C-Format

The Mathematics of the SEAM-LTM UJ Plugin

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Abstract

This document derives the mathematics of the `x2uhj` plugin step by step. It covers the AmbiX-to-UJ C-format encoding, the origin of Gerzon’s matrix coefficients, and the sample-rate-independent design of the quadrature all-pass network.

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1 Introduction and Historical Framing

UJ is a hierarchical multichannel encoding system that represents a periphonic or horizontal sound field in a small number of channels suitable for stereo-compatible transmission and subsequent decoding. The name “UJ” derives from the BBC Matrix-H system and the 45J variant, both developed at the BBC in the 1970s as practical approaches to compatible surround sound. Michael Gerzon unified and extended these systems in his 1985 AES paper, establishing the coefficient set that the `x2uhj` plugin implements. The C-format variant carries four channels: the stereo pair (L, R) , a third directional channel T , and a height-related channel Q .

The `x2uhj` plugin accepts a First-Order AmbiX signal (four channels in ACN order, SN3D normalisation) and delivers UJ C-format output. This document derives Gerzon’s matrix coefficients by inverse verification against known decoded loudspeaker positions. It establishes the quadrature all-pass network as a sample-rate-independent design procedure, enabling consistent behaviour across all standard audio sample rates.

2 AmbiX to FuMa

The plugin first converts the incoming ACN/SN3D channels to the FuMa convention used throughout the UJ literature. ACN orders the four first-order channels as A_0 (omnidirectional), A_1 (front-back), A_2 (up-down), A_3 (left-right), corresponding to the spherical harmonic sequence W, Y, Z, X . FuMa arranges the same signals as W, X, Y, Z and scales the omnidirectional component by $1/\sqrt{2}$. The conversion is therefore a reindexing together with a single rescaling:

$$W = \frac{1}{\sqrt{2}} A_0, \quad X = A_3, \quad Y = A_1, \quad Z = A_2. \quad (1)$$

All subsequent matrix operations in this document refer to the FuMa signals W, X, Y, Z .

3 The UHJ C-Format Matrix

Gerzon's UHJ encoding proceeds in two stages: formation of the intermediate sum and difference signals Σ and Δ , followed by the channel assignments for L, R, T , and Q .

$$\Sigma = 0.9396926 W + 0.1855740 X, \quad (2)$$

$$\Delta = j(-0.3420201 W + 0.5098604 X) + 0.6554516 Y, \quad (3)$$

$$L = \frac{1}{2}(\Sigma + \Delta), \quad R = \frac{1}{2}(\Sigma - \Delta), \quad (4)$$

$$T = j(-0.1432 W + 0.6512 X) - 0.7071 Y, \quad Q = 0.9772 Z. \quad (5)$$

Σ is the forward-facing sum channel; its first-order pattern secures mono compatibility across all playback configurations. Δ is the difference channel; the factor j introduces a broadband 90-degree phase shift that encodes lateral width. L and R are the two transmitted stereo channels, formed from the sum and difference of Σ and Δ . T is the third C-format channel; it carries additional directional resolution that extends spatial accuracy beyond the stereo base. Q is the height-related channel; it carries the vertical component Z scaled by 0.9772 to match the overall normalisation scheme. The symbol j denotes a broadband 90-degree phase shift applied across the full audio bandwidth; its implementation as a sample-rate-independent all-pass network is the subject of Sections 5 and 6.

4 Reconstructing Gerzon's Coefficients

Gerzon chose the UHJ coefficients to reproduce the psychoacoustic localization of the original sound field. The velocity vector $\mathbf{r}_V$ governs perceived direction below roughly 700 Hz, and the energy vector $\mathbf{r}_E$ governs it above. Both vectors are weighted averages of the loudspeaker direction unit vectors $\hat{\mathbf{u}}_i$:

$$\mathbf{r}_V = \frac{\sum_i g_i \hat{\mathbf{u}}_i}{\sum_i g_i}, \quad \mathbf{r}_E = \frac{\sum_i g_i^2 \hat{\mathbf{u}}_i}{\sum_i g_i^2}. \quad (6)$$

The velocity vector uses the loudspeaker gains g_i directly, and the energy vector uses their squared values. A well-designed encoding keeps both vectors pointing toward the original source azimuth, with magnitude near unity.

The constants in the Σ and Δ expressions carry geometric meaning. The coefficient $0.9396926 = \cos 20^\circ$ and $0.3420201 = \sin 20^\circ$ reveal the 20° design angle that Gerzon selected to balance spatial precision and robust mono compatibility. Σ realises a forward first-order microphone pattern, which secures mono compatibility across all playback configurations. The j term in Δ supplies the degree of freedom that maps the full 360° of azimuth into two channels reversibly.

Inverse verification takes the published coefficients as given and confirms numerically that they realise good localization. The verification evaluates two listening models. The surround model feeds (L, R) to a symmetric illustrative loudspeaker layout and reads the localization vectors of the reconstructed field. The super-stereo model feeds (L, R) to two frontal loudspeakers at 30° and -30° and reads the localization a listener perceives at those positions.

The surround model illuminates why Gerzon selected these coefficients: the reconstructed field localises faithfully across the full azimuth circle. The super-stereo model demonstrates the stereo listening result: a frontal pair at $\pm 30^\circ$ conveys the same source direction with small angular error. Agreement between the two models across azimuth confirms that the published coefficients serve both listening contexts.

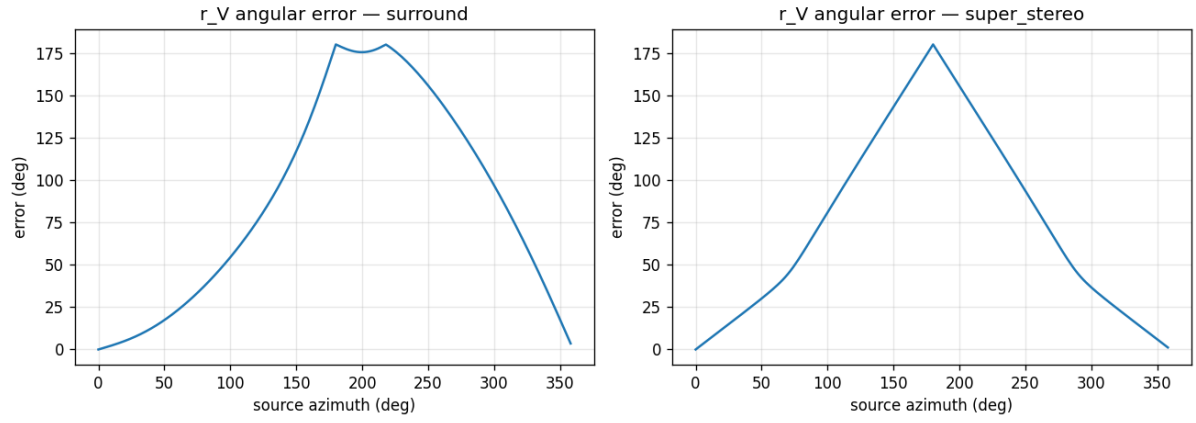


Figure 1: Velocity-vector angular error versus source azimuth: surround decode (left) and super-stereo (right). Both models confirm that Gerzon’s published coefficients localise the source across azimuth.

References